msvtx_el3hcal

MC vs data · overview · rank 59 / 98

rank 59 / 98 by endcap D_equal

used by: NN1 barrel, NN1 endcap

rank-identical columns merged into this one:
    summedCaloClusters/el3hcal
    MS1Vtx_l3hcal_raw
    MS1Vtx_barrel_l3hcal
    MS1Vtx_endcap_l3hcal

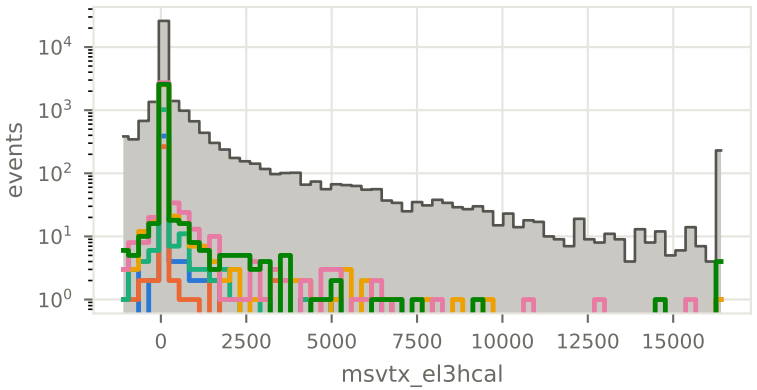
region	D_equal	noise	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg	rank
endcap	0.193	0.022	0.192	0.196	0.190	0.196	0.197	0.184	0.196	34,970	59/98
barrel	0.240	0.051	0.236	0.265	0.235	0.236	0.245	0.239	0.224	1,089	53/98
all	0.178	0.018	0.176	0.190	0.174	0.181	0.183	0.170	0.172	36,059	77/98

msvtx_el3hcal

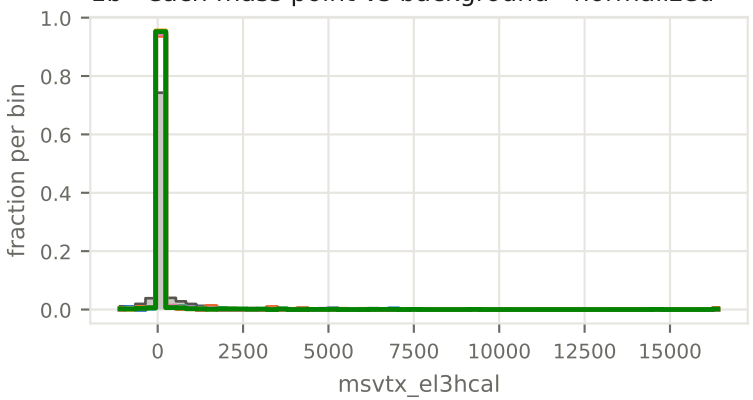
region: ENDCAP · rank 59 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.193 (noise 0.022) D_pooled 0.192 min 0.184 (mH600_mS150) max 0.197 (mH400_mS100) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

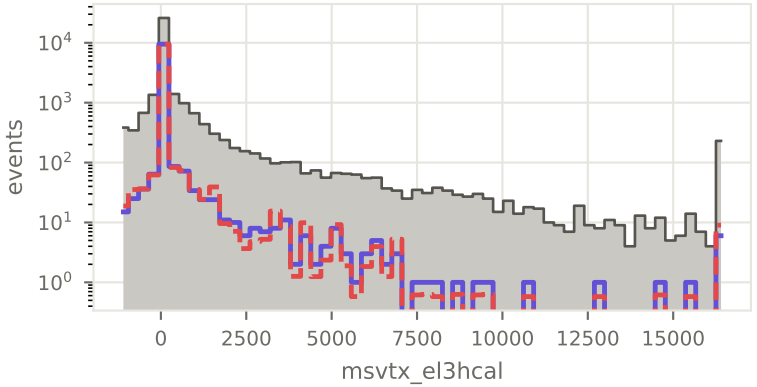


1b · each mass point vs background · normalized

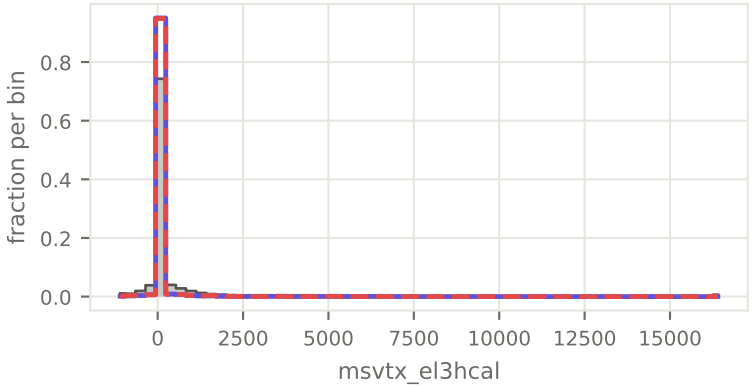


1 · each mass point
background n 34,970
mH125_mS15 D 0.196 n 411
mH125_mS55 D 0.190 n 281
mH200_mS50 D 0.196 n 1,071
mH400_mS100 D 0.197 n 2,648
mH600_mS150 D 0.184 n 2,882
mH1000_mS275 D 0.196 n 2,703

2a · combined signal vs background · raw

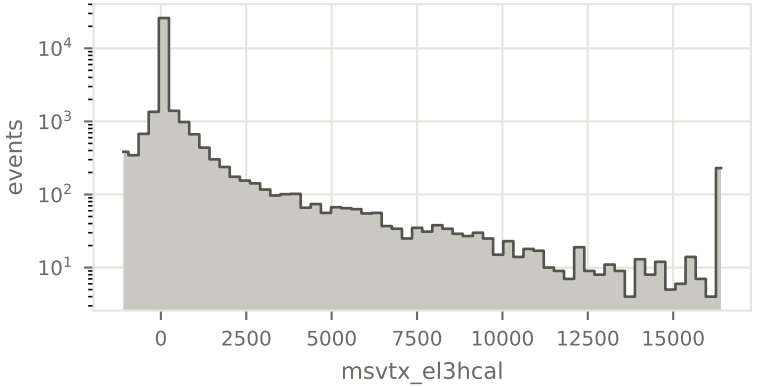


2b · combined signal vs background · normalized

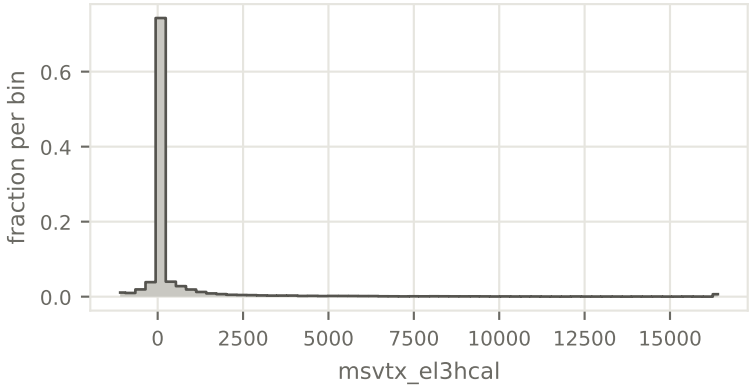


2 · combined signal
background n 34,970
combined, pooled D 0.192 n 9,996
equal weight per mass D 0.193

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

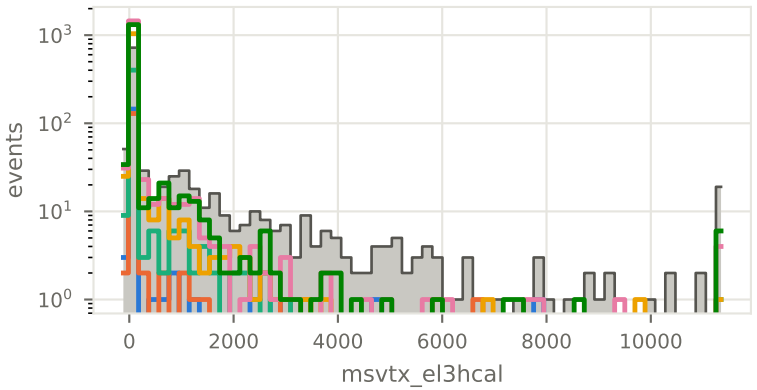
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el3hcal

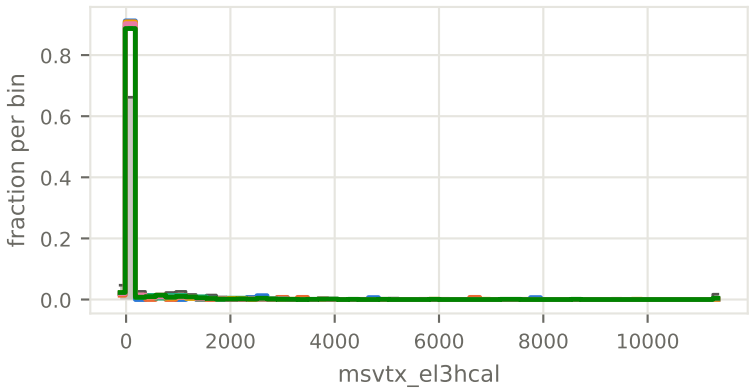
region: BARREL · rank 59 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.240 (noise 0.051) D_pooled 0.236 min 0.224 (mH1000_mS275) max 0.265 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

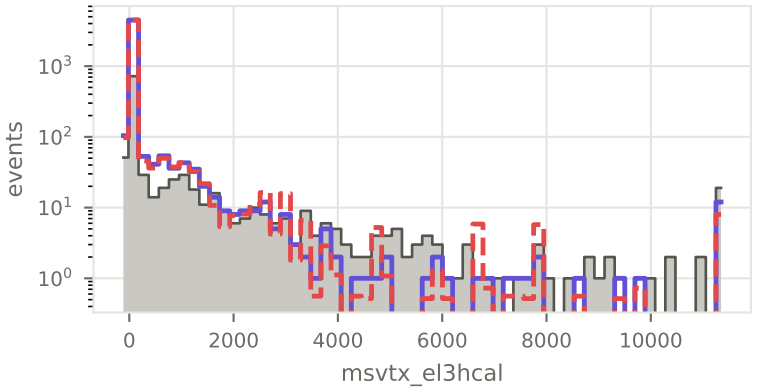


1b · each mass point vs background · normalized

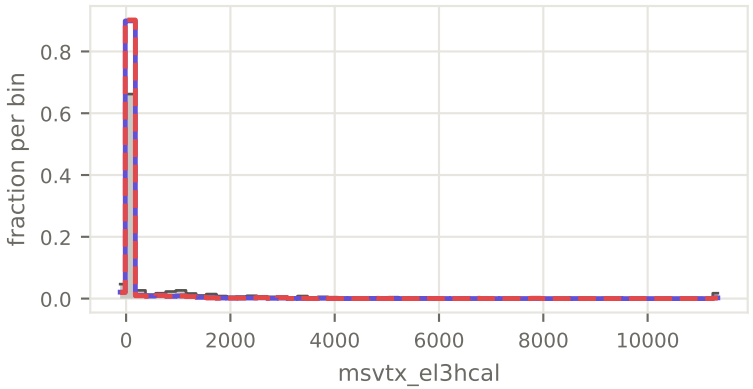


1 · each mass point
background n 1,089
mH125_mS15 D 0.265 n 159
mH125_mS55 D 0.235 n 142
mH200_mS50 D 0.236 n 449
mH400_mS100 D 0.245 n 1,144
mH600_mS150 D 0.239 n 1,608
mH1000_mS275 D 0.224 n 1,485

2a · combined signal vs background · raw

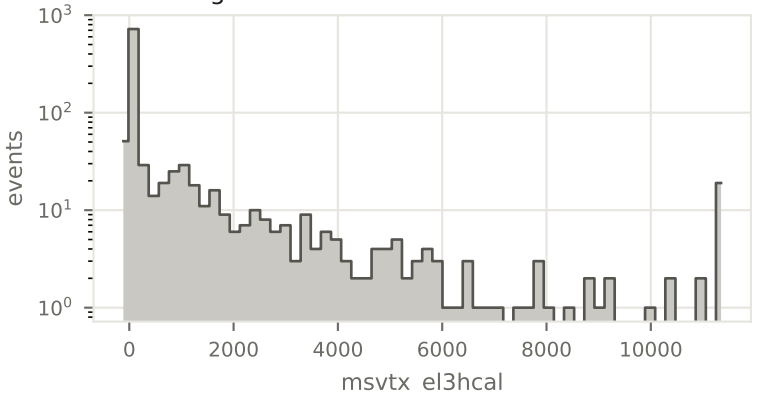


2b · combined signal vs background · normalized

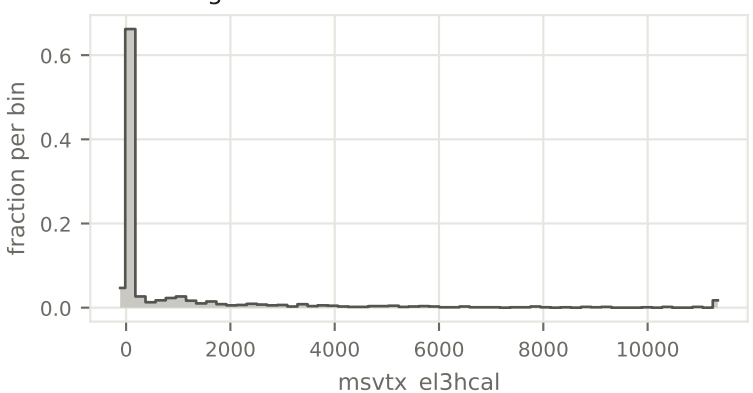


2 · combined signal
background n 1,089
combined, pooled D 0.236 n 4,987
equal weight per mass D 0.240

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

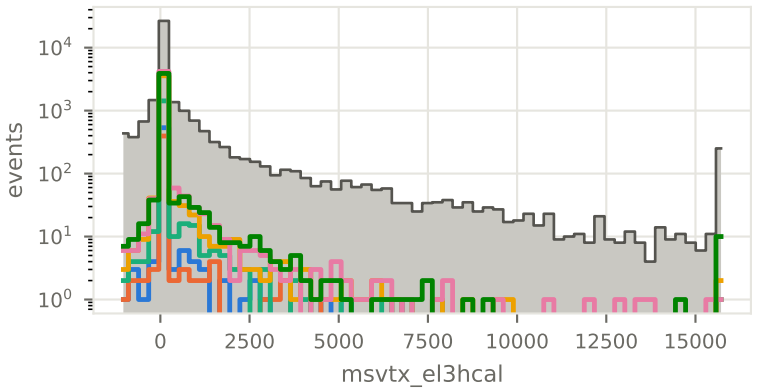
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el3hcal

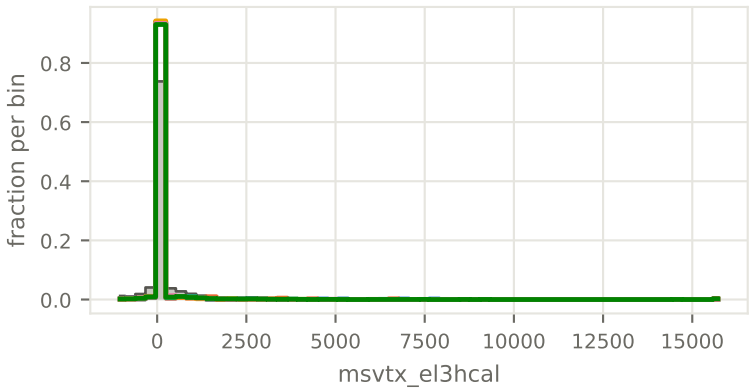
region: ALL · rank 59 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.178 (noise 0.018) D_pooled 0.176 min 0.170 (mH600_mS150) max 0.190 (mH125_mS15) bkg n = 36,059 sig n = 14,983

1a · each mass point vs background · raw

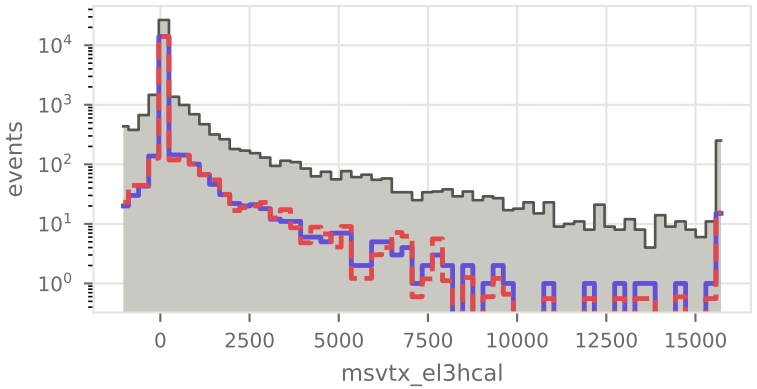


1b · each mass point vs background · normalized

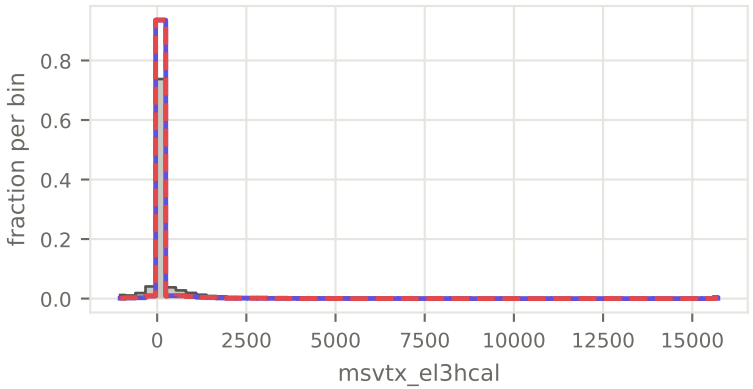


1 · each mass point
background n 36,059
mH125_mS15 D 0.190 n 570
mH125_mS55 D 0.174 n 423
mH200_mS50 D 0.181 n 1,520
mH400_mS100 D 0.183 n 3,792
mH600_mS150 D 0.170 n 4,490
mH1000_mS275 D 0.172 n 4,188

2a · combined signal vs background · raw

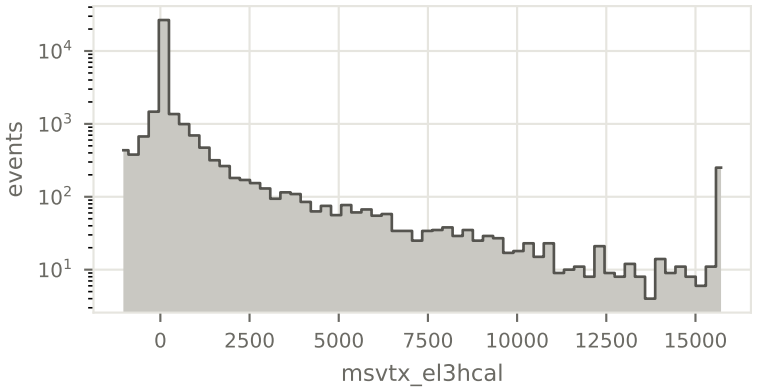


2b · combined signal vs background · normalized

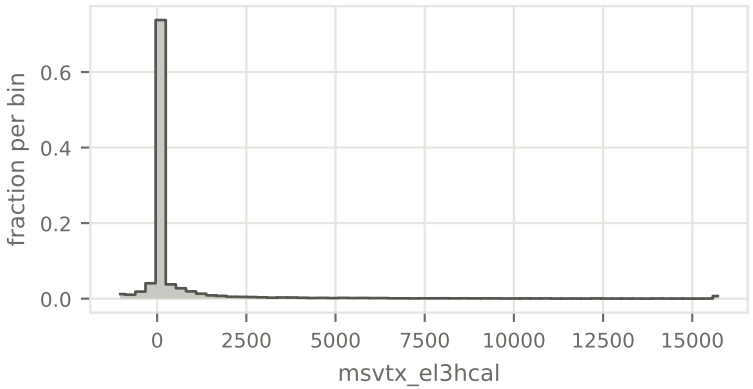


2 · combined signal
background n 36,059
combined, pooled D 0.176 n 14,983
equal weight per mass D 0.178

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed